Ppk能力值汇总

客户	联合电子UAES	零件号	F01R20Y574-1#	零件名称	阀体	日期	2025/7/21

序号	尺寸编号	名称	规格			客户要求能力值	实测能力值	是否通过	
			spec.	upper Ltd.	lower Ltd.	Ppk	Ppk	Y/N	
1	113#	螺钉孔大端	3.8	0	-0.1	>1.67	**1.978**	Y	
2	113-1#	螺钉孔大端	3.8	0	-0.1	>1.67	**3.567**	Y	
3	119#	螺钉孔小端	3.5	0	-0.1	>1.67	**3.815**	Y	
4	119-1#	螺钉孔小端	3.5	0	-0.1	>1.67	**4.574**	Y	
5	145#	H面平面度	0.15	0.15	0	>1.67	**2.032**	Y	
6	267-1#	卡位厚度	6.6	0.2	-0.1	>1.67	**3.154**	Y	
7	267-2#	卡位厚度	6.6	0.2	-0.1	>1.67	**1.807**	Y	
8	267-3#	卡位厚度	6.6	0.2	-0.1	>1.67	**1.754**	Y	
9	267-4#	卡位厚度	6.6	0.2	-0.1	>1.67	**4.547**	Y	
10	267-5#	卡位厚度	6.6	0.2	-0.1	>1.67	**2.969**	Y	
11	267-6#	卡位厚度	6.6	0.2	-0.1	>1.67	**2.945**	Y	

精明（西安）金属科技有限公司

CHING MING METAL & TECHNOLOGY CO.,LTD

NO: 20250721001 DATE：25-7-21

spec	3.8	
upper tol.	0.0	3.800
lower tol.	-0.1	3.700

CUSTOMER: UAES
Part Name: Throttle Housing
Part NO:F01R 20Y 574

Instrument: CMM
Inspector:Liu yajuan
Drawing item:113#

CP= 3.519
Ca= 0.434
Validation:

CPK= 1.994
PP= 3.491
Approval：

PpK= 1.978

				σ_S	0.0048
		Max	3.7799	σ_R	0.0047
Tmax-Tmin	0.0110	Min	3.7632	average	3.7717

team	No. 1	No. 2	No. 3	No. 4	No. 5	No. 6	No. 7	No. 8	No. 9	No. 10	No. 11	No. 12	No. 13	No. 14	No. 15	No. 16	No. 17	No. 18	No. 19	No. 20	No. 21	No. 22	No. 23	No. 24	No. 25	average
sample 1	3.7792	3.7690	3.7753	3.7713	3.7751	3.7635	3.7735	3.7684	3.7649	3.7659																3.7706
sample 2	3.7724	3.7723	3.7781	3.7771	3.7653	3.7646	3.7772	3.7641	3.7694	3.7746																3.7715
sample 3	3.7782	3.7799	3.7786	3.7769	3.7660	3.7781	3.7768	3.7704	3.7736	3.7707																3.7749
sample 4	3.7654	3.7700	3.7716	3.7740	3.7635	3.7690	3.7693	3.7632	3.7710	3.7728																3.7690
sample 5	3.7713	3.7703	3.7724	3.7681	3.7780	3.7699	3.7687	3.7736	3.7776	3.7740																3.7724
sample 6	3.7733	3.7723	3.7752	3.7735	3.7696	3.7690	3.7731	3.7679	3.7713	3.7716																3.7717
sample 7	0.0138	0.0109	0.0069	0.0090	0.0145	0.0146	0.0086	0.0105	0.0126	0.0087																0.01101

7.000 0.577 d2= 2.326 D3= 0.000 D4= 2.114 $UCL_R = D_4\bar{R} =$ 0.02329 $LCL_R = D_3\bar{R} =$ 0 $UCL_X = \bar{\bar{X}} + A_2\bar{R} =$ 3.77803 $LCL_X = \bar{\bar{X}} - A_2\bar{R} =$ 3.76532

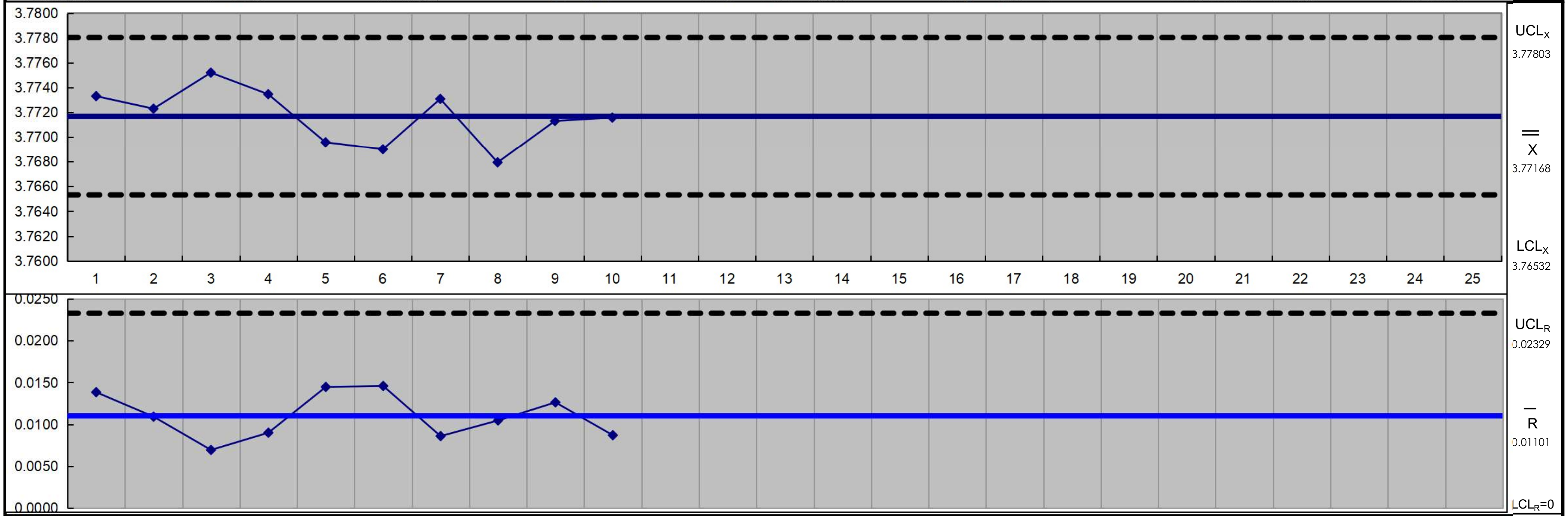


chart analysis	a)Note：	R chart and $\bar{X}$ chart should be analysed seperately, R chart first，then $\bar{X}$ chart.
	b) out of limit：	point out of upper or lower limit, include point on the limit line.
	c)Chain：	1> 7 continuous points in one side of average line.
	d) clane：	7 continuous points with trend, go up or go down.
	e)Close：	1> about 2/3 points in 1/3 limits lines area. 2> about 2/3 points close to average line.

unnormal point analyse and improvement method	

CHING MING METAL & TECHNOLOGY CO.,LTD

NO: 20250721001

DATE：25-7-21

spec	3.8	
upper tol.	0.0	3.800
lower tol.	-0.1	3.700

CUSTOMER: UAES
Part Name: Throttle Housing
Part NO:F01R 20Y 574

Instrument: CMM
Inspector:Liu yajuan
Drawing item:113-1#

CP= 4.042
Ca= 0.124
Validation:

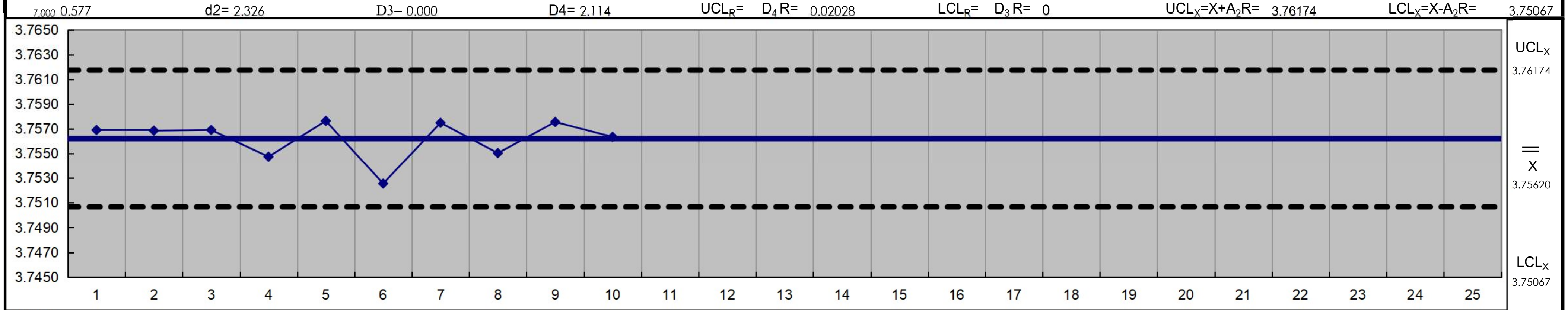
CPK= 3.541
PP= 4.072
Approval：

PpK= 3.567

				σ_S	0.0041
		Max	3.7629	σ_R	0.0041
Tmax-Tmin	0.0096	Min	3.7486	average	3.7562

team	No. 1	No. 2	No. 3	No. 4	No. 5	No. 6	No. 7	No. 8	No. 9	No. 10	No. 11	No. 12	No. 13	No. 14	No. 15	No. 16	No. 17	No. 18	No. 19	No. 20	No. 21	No. 22	No. 23	No. 24	No. 25	average
sample 1	3.7625	3.7555	3.7533	3.7530	3.7629	3.7540	3.7506	3.7516	3.7566	3.7540																3.7554
sample 2	3.7547	3.7572	3.7624	3.7522	3.7486	3.7546	3.7615	3.7614	3.7616	3.7524																3.7567
sample 3	3.7533	3.7553	3.7566	3.7625	3.7565	3.7516	3.7580	3.7627	3.7565	3.7557																3.7569
sample 4	3.7556	3.7567	3.7526	3.7559	3.7583	3.7487	3.7612	3.7506	3.7546	3.7586																3.7553
sample 5	3.7583	3.7595	3.7596	3.7502	3.7620	3.7539	3.7561	3.7490	3.7585	3.7610																3.7568
sample 6	3.7569	3.7569	3.7569	3.7547	3.7576	3.7526	3.7575	3.7550	3.7576	3.7563																3.7562
sample 7	0.0092	0.0042	0.0098	0.0124	0.0143	0.0059	0.0109	0.0137	0.0070	0.0086																0.00959

7.000 0.577   d2= 2.326   D3= 0.000   D4= 2.114 $UCL_R = \overline{D_4 R} =$ 0.02028 $LCL_R = \overline{D_3 R} =$ 0 $UCL_X = \overline{\overline{X}} + A_2\overline{R} =$ 3.76174 $LCL_X = \overline{\overline{X}} - A_2\overline{R} =$ 3.75067



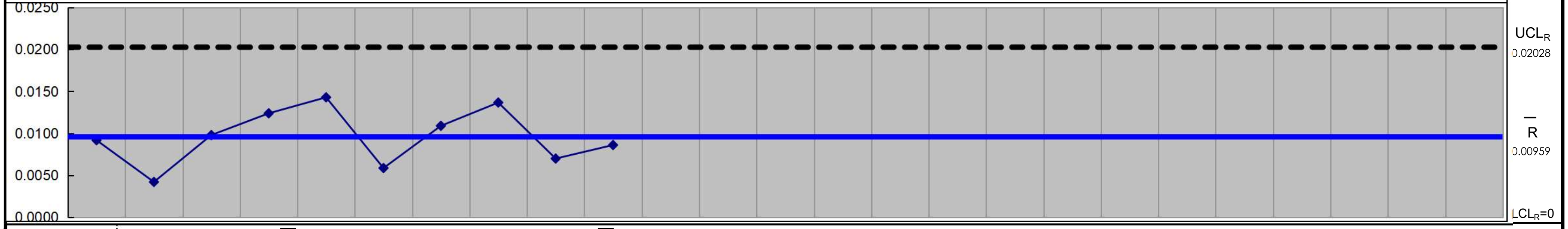


chart analysis		
	a)Note：	R chart and $\overline{X}$ chart should be analysed seperately, R chart first，then $\overline{X}$ chart.
	b) out of limit：	point out of upper or lower limit, include point on the limit line.
	c)Chain：	1> 7 continuous points in one side of average line.
	d) clane：	7 continuous points with trend, go up or go down.
	e)Close：	1> about 2/3 points in 1/3 limits lines area. 2> about 2/3 points close to average line.

unnormal point analyse and improvement method

CHING MING METAL & TECHNOLOGY CO.,LTD

NO: 20250721001 DATE：25-7-21

CUSTOMER: UAES
Part Name: Throttle Housing
Part NO:F01R 20Y 574

Instrument: CMM
Inspector:Liu yajuan
Drawing item:119#

CP= 4.264
Ca= 0.113
Validation:

CPK= 3.783
PP= 4.301
Approval：

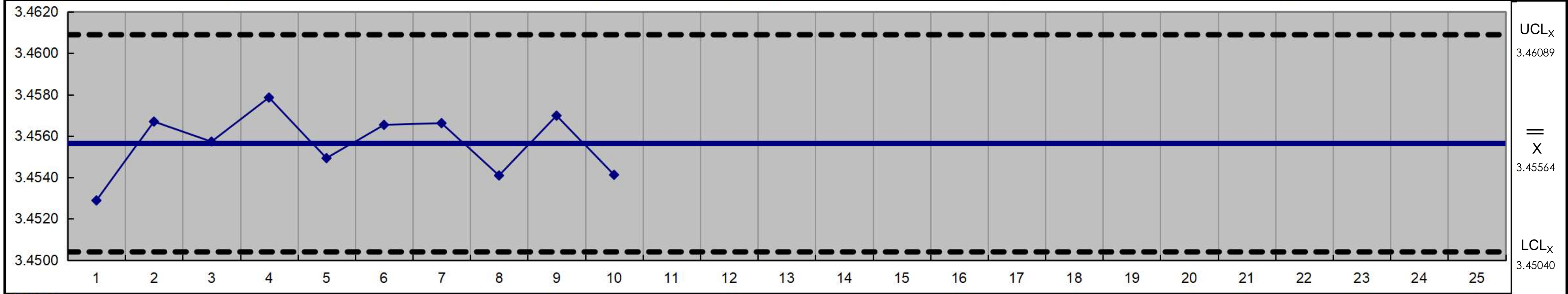
PpK= 3.815

spec	3.5	
upper tol.	0.0	3.500
lower tol.	-0.1	3.400

				σ_S	0.0039
		Max	3.4615	σ_R	0.0039
Tmax-Tmin	0.0091	Min	3.4486	average	3.4556

team	No. 1	No. 2	No. 3	No. 4	No. 5	No. 6	No. 7	No. 8	No. 9	No. 10	No. 11	No. 12	No. 13	No. 14	No. 15	No. 16	No. 17	No. 18	No. 19	No. 20	No. 21	No. 22	No. 23	No. 24	No. 25	average
sample 1	3.4572	3.4592	3.4588	3.4613	3.4576	3.4586	3.4590	3.4556	3.4587	3.4541																3.4580
sample 2	3.4561	3.4496	3.4513	3.4520	3.4532	3.4539	3.4490	3.4560	3.4495	3.4606																3.4531
sample 3	3.4487	3.4552	3.4523	3.4551	3.4502	3.4583	3.4562	3.4531	3.4579	3.4502																3.4537
sample 4	3.4538	3.4595	3.4548	3.4601	3.4565	3.4544	3.4594	3.4486	3.4588	3.4497																3.4556
sample 5	3.4486	3.4599	3.4615	3.4607	3.4570	3.4576	3.4596	3.4570	3.4600	3.4560																3.4578
sample 6	3.4529	3.4567	3.4557	3.4579	3.4549	3.4565	3.4566	3.4541	3.4570	3.4541																3.4556
sample 7	0.0086	0.0102	0.0102	0.0093	0.0074	0.0047	0.0106	0.0084	0.0106	0.0109																0.00909

7.000 0.577 d2= 2.326 D3= 0.000 D4= 2.114 $UCL_R=$ $D_4\bar{R}=$ 0.01922 $LCL_R=$ $D_3\bar{R}=$ 0 $UCL_X=\bar{\bar{X}}+A_2\bar{R}=$ 3.46089 $LCL_X=\bar{\bar{X}}-A_2\bar{R}=$ 3.45040



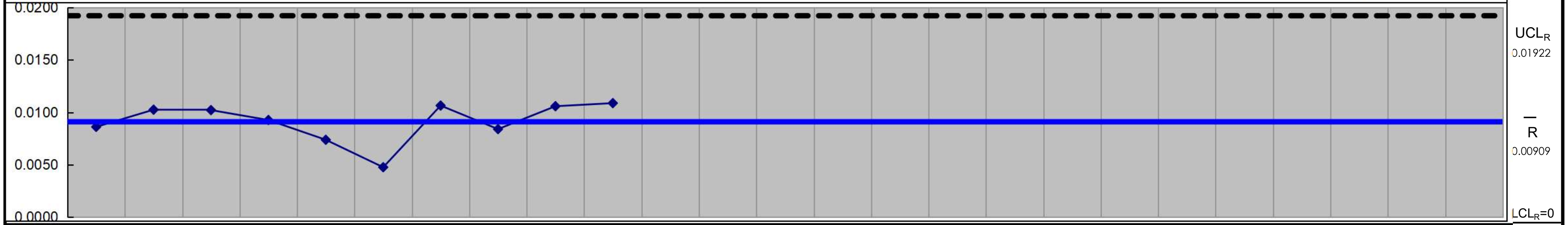


chart analysis	a)Note：	R chart and X chart should be analysed seperately, R chart first，then X chart.	b) out of limit：	point out of upper or lower limit, include point on the limit line.
	c)Chain：	1> 7 continuous points in one side of average line.	d) clane：	7 continuous points with trend, go up or go down.
	e)Close：	1> about 2/3 points in 1/3 limits lines area. 2> about 2/3 points close to average line.		

unnormal point analyse and improvement method	

CHING MING METAL & TECHNOLOGY CO.,LTD

NO: 20250721001 DATE：25-7-21

spec	3.5	
upper tol.	0.0	3.500
lower tol.	-0.1	3.400

CUSTOMER: UAES
Part Name: Throttle Housing
Part NO:F01R 20Y 574

Instrument: CMM
Inspector:Liu yajuan
Drawing item:119-1#

CP= 4.982
Ca= 0.056
Validation:

CPK= 4.702
PP= 4.847
Approval：

PpK= 4.574

				σ_S	0.0034
		Max	3.4599	σ_R	0.0033
Tmax-Tmin	0.0078	Min	3.4481	average	3.4528

team	No. 1	No. 2	No. 3	No. 4	No. 5	No. 6	No. 7	No. 8	No. 9	No. 10	No. 11	No. 12	No. 13	No. 14	No. 15	No. 16	No. 17	No. 18	No. 19	No. 20	No. 21	No. 22	No. 23	No. 24	No. 25	average
sample 1	3.4592	3.4502	3.4515	3.4526	3.4504	3.4554	3.4499	3.4555	3.4587	3.4519																3.4535
sample 2	3.4502	3.4485	3.4494	3.4498	3.4527	3.4530	3.4592	3.4496	3.4534	3.4598																3.4526
sample 3	3.4487	3.4498	3.4501	3.4565	3.4502	3.4514	3.4522	3.4493	3.4503	3.4599																3.4518
sample 4	3.4500	3.4547	3.4531	3.4529	3.4509	3.4532	3.4481	3.4526	3.4517	3.4543																3.4522
sample 5	3.4548	3.4591	3.4506	3.4524	3.4527	3.4571	3.4561	3.4492	3.4595	3.4483																3.4540
sample 6	3.4526	3.4525	3.4510	3.4528	3.4514	3.4540	3.4531	3.4512	3.4547	3.4549																3.4528
sample 7	0.0105	0.0105	0.0037	0.0067	0.0026	0.0056	0.0111	0.0064	0.0092	0.0115																0.00778

7.000 0.577 d2= 2.326 D3= 0.000 D4= 2.114 $UCL_R= D_4\bar{R}=$ 0.01645 $LCL_R= D_3\bar{R}=$ 0 $UCL_X=\bar{\bar{X}}+A_2\bar{R}=$ 3.45730 $LCL_X=\bar{\bar{X}}-A_2\bar{R}=$ 3.44832

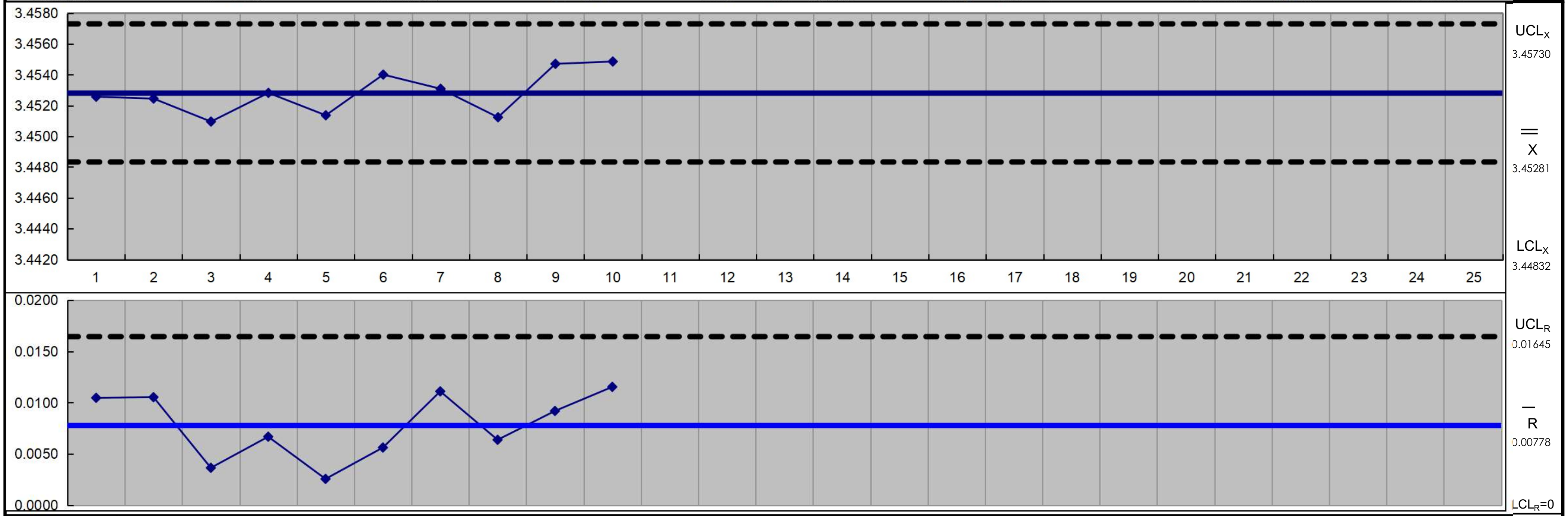


chart analysis

a)Note： $\bar{R}$ chart and $\bar{X}$ chart should be analysed seperately, $\bar{R}$ chart first，then X chart.

b) out of limit： point out of upper or lower limit, include point on the limit line.

c)Chain： 1> 7 continuous points in one side of average line.

d) clane： 7 continuous points with trend, go up or go down.

e)Close： 1> about 2/3 points in 1/3 limits lines area. 2> about 2/3 points close to average line.

unnormal point analyse and improvement method

CHING MING METAL & TECHNOLOGY CO.,LTD

NO: 20250721001
DATE：25-7-21

spec	0.000	
upper tol.	0.150	0.150
lower tol.	0.000	0.000

CUSTOMER: UAES
Part Name: Throttle Housing
Part NO:F01R 20Y 574

Instrument: CMM
Inspector:Liu yajuan
Drawing item:145#

CP= 3.256
Ca= -0.356
Validation:

CPK= 2.096
PP= 3.157
Approval：

PpK= 2.032

				σ_S	0.0079
		Max	0.0630	σ_R	0.0077
Tmax-Tmin	0.0179	Min	0.0354	average	0.0483

team	No. 1	No. 2	No. 3	No. 4	No. 5	No. 6	No. 7	No. 8	No. 9	No. 10	No. 11	No. 12	No. 13	No. 14	No. 15	No. 16	No. 17	No. 18	No. 19	No. 20	No. 21	No. 22	No. 23	No. 24	No. 25	average
sample 1	0.0460	0.0354	0.0481	0.0472	0.0509	0.0439	0.0400	0.0601	0.0619	0.0445																0.0478
sample 2	0.0380	0.0412	0.0579	0.0422	0.0474	0.0406	0.0462	0.0528	0.0548	0.0362																0.0457
sample 3	0.0403	0.0504	0.0588	0.0374	0.0630	0.0522	0.0421	0.0501	0.0565	0.0568																0.0508
sample 4	0.0607	0.0401	0.0422	0.0565	0.0592	0.0572	0.0373	0.0488	0.0425	0.0611																0.0506
sample 5	0.0461	0.0366	0.0522	0.0484	0.0578	0.0444	0.0441	0.0403	0.0453	0.0499																0.0465
sample 6	0.0462	0.0407	0.0519	0.0464	0.0556	0.0477	0.0419	0.0504	0.0522	0.0497																0.0483
sample 7	0.0226	0.0150	0.0167	0.0190	0.0157	0.0166	0.0088	0.0198	0.0193	0.0249																0.01786

7.000 0.577　　d2= 2.326　　D3= 0.000　　D4= 2.114　　$UCL_R = D_4\bar{R} =$ 0.03776　　$LCL_R = D_3\bar{R} =$ 0　　$UCL_X = \bar{\bar{X}} + A_2\bar{R} =$ 0.05858　　$LCL_X = \bar{\bar{X}} - A_2\bar{R} =$ 0.03797

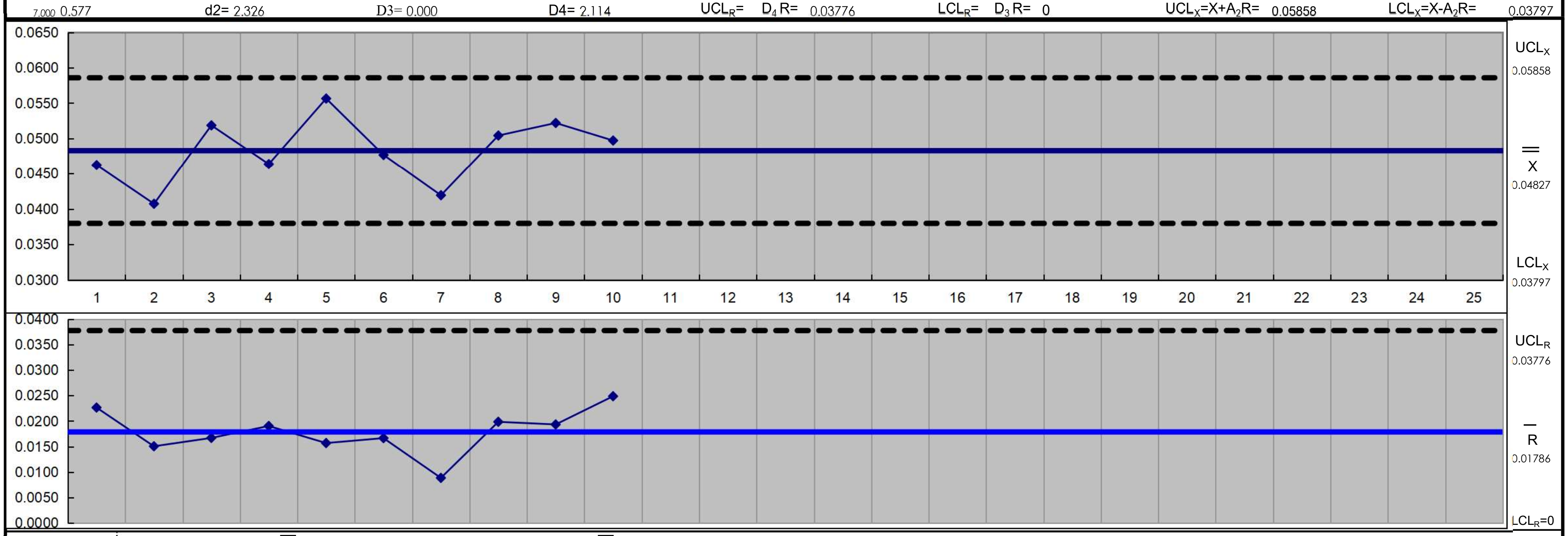


chart analysis	a)Note：	R chart and X chart should be analysed seperately, R chart first，then X chart.
		b) out of limit： point out of upper or lower limit, include point on the limit line.
	c)Chain：	1> 7 continuous points in one side of average line.
		d) clane： 7 continuous points with trend, go up or go down.
	e)Close：	1> about 2/3 points in 1/3 limits lines area. 2> about 2/3 points close to average line.

unnormal point analyse and improvement method	

CHING MING METAL & TECHNOLOGY CO.,LTD

NO: 20250721001

DATE：25-7-21

spec	6.6	
upper tol.	0.2	6.800
lower tol.	-0.1	6.500

CUSTOMER: UAES
Part Name: Throttle Housing
Part NO:F01R 20Y 574

Instrument: C
Inspector:Liu yajuan
Drawing item:267-1#

CP= 4.529
Ca= -0.350
Validation:

CPK= 2.945
PP= 4.851
Approval：

PpK= 3.154

		σ_S	0.01
Max	6.61	σ_R	0.01
Min	6.58	average	6.60

Tmax-Tmin 0.0257

team	No. 1	No. 2	No. 3	No. 4	No. 5	No. 6	No. 7	No. 8	No. 9	No. 10	No. 11	No. 12	No. 13	No. 14	No. 15	No. 16	No. 17	No. 18	No. 19	No. 20	No. 21	No. 22	No. 23	No. 24	No. 25	average
sample 1	6.61	6.60	6.58	6.61	6.60	6.61	6.59	6.60	6.60	6.59																6.60
sample 2	6.58	6.58	6.61	6.59	6.58	6.59	6.60	6.59	6.58	6.59																6.59
sample 3	6.61	6.59	6.60	6.59	6.60	6.60	6.61	6.60	6.59	6.58																6.60
sample 4	6.60	6.60	6.59	6.61	6.60	6.61	6.59	6.61	6.61	6.61																6.60
sample 5	6.61	6.59	6.60	6.61	6.59	6.60	6.60	6.58	6.58	6.60																6.60
sample 6	6.605	6.593	6.595	6.601	6.595	6.603	6.597	6.599	6.593	6.595																6.598
sample 7	0.031	0.022	0.029	0.020	0.022	0.023	0.020	0.032	0.027	0.030																0.026

7.000 0.577　d2= 2.326　D3= 0.000　D4= 2.114　$UCL_R= D_4\bar{R}=$ 0.05428　$LCL_R= D_3\bar{R}=$ 0　$UCL_X=\bar{\bar{X}}+A_2\bar{R}=$ 6.61233　$LCL_X=\bar{\bar{X}}-A_2\bar{R}=$ 6.58270

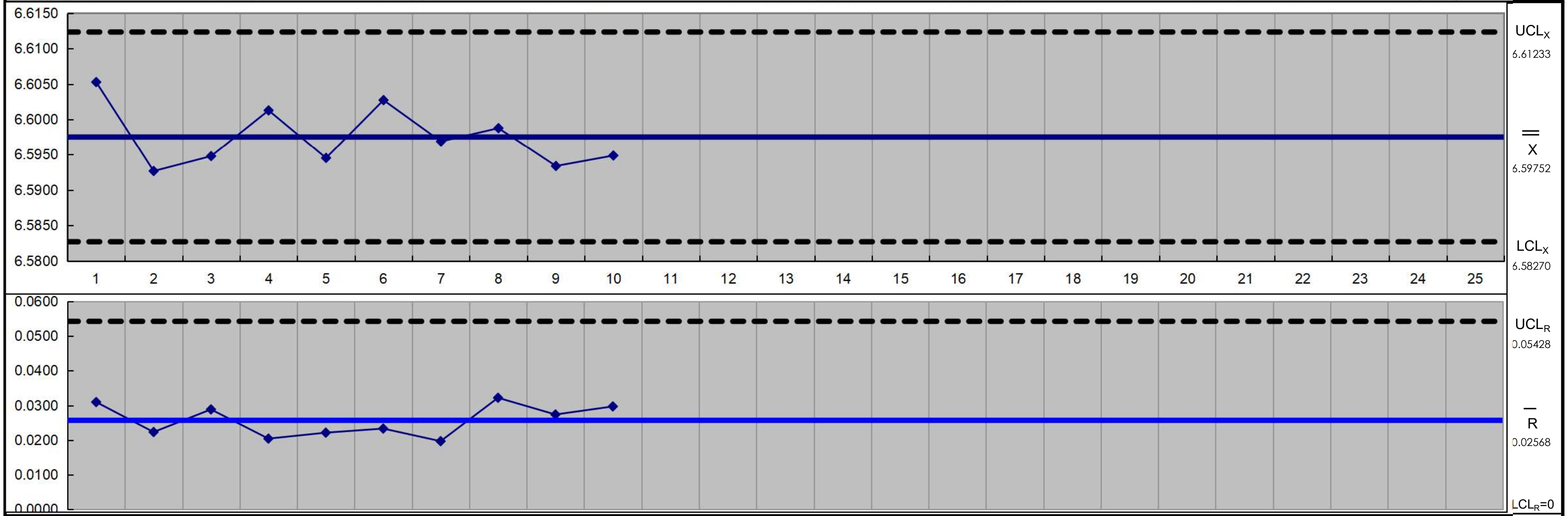


chart analysis	a)Note：	R chart and $\bar{X}$ chart should be analysed seperately, R chart first，then $\bar{X}$ chart. b) out of limit： point out of upper or lower limit, include point on the limit line.
	c)Chain：	1> 7 continuous points in one side of average line. d) clane： 7 continuous points with trend, go up or go down.
	e)Close：	1> about 2/3 points in 1/3 limits lines area. 2> about 2/3 points close to average line.

unnormal point analyse and improvement method

CHING MING METAL & TECHNOLOGY CO.,LTD

NO: 20250721001 DATE：25-7-21

spec	6.6	
upper tol.	0.2	6.800
lower tol.	-0.1	6.500

CUSTOMER: UAES
Part Name: Throttle Housing
Part NO:F01R 20Y 574

Instrument: C
Inspector:Liu yajuan
Drawing item:267-2#

CP= 4.674
Ca= -0.616
Validation:

CPK= 1.794
PP= 4.708
Approval：

PpK= 1.807

				σ_S	0.01
		Max	6.57	σ_R	0.01
Tmax-Tmin	0.0249	Min	6.54	average	6.56

team	No. 1	No. 2	No. 3	No. 4	No. 5	No. 6	No. 7	No. 8	No. 9	No. 10	No. 11	No. 12	No. 13	No. 14	No. 15	No. 16	No. 17	No. 18	No. 19	No. 20	No. 21	No. 22	No. 23	No. 24	No. 25	average
sample 1	6.56	6.57	6.55	6.57	6.55	6.56	6.56	6.54	6.57	6.57																6.56
sample 2	6.57	6.54	6.54	6.54	6.55	6.56	6.54	6.54	6.56	6.56																6.55
sample 3	6.56	6.54	6.55	6.57	6.56	6.55	6.56	6.55	6.56	6.54																6.55
sample 4	6.55	6.57	6.56	6.57	6.54	6.55	6.57	6.57	6.55	6.54																6.56
sample 5	6.56	6.57	6.57	6.55	6.57	6.56	6.56	6.57	6.56	6.56																6.56
sample 6	6.559	6.560	6.555	6.562	6.554	6.558	6.558	6.553	6.560	6.557																6.558
sample 7	0.027	0.033	0.029	0.033	0.024	0.009	0.028	0.024	0.014	0.028																0.025

7.000 0.577 d2= 2.326 D3= 0.000 D4= 2.114 $UCL_R= D_4\bar{R}=$ 0.05260 $LCL_R= D_3\bar{R}=$ 0 $UCL_X=\bar{\bar{X}}+A_2\bar{R}=$ 6.57192 $LCL_X=\bar{\bar{X}}-A_2\bar{R}=$ 6.54321

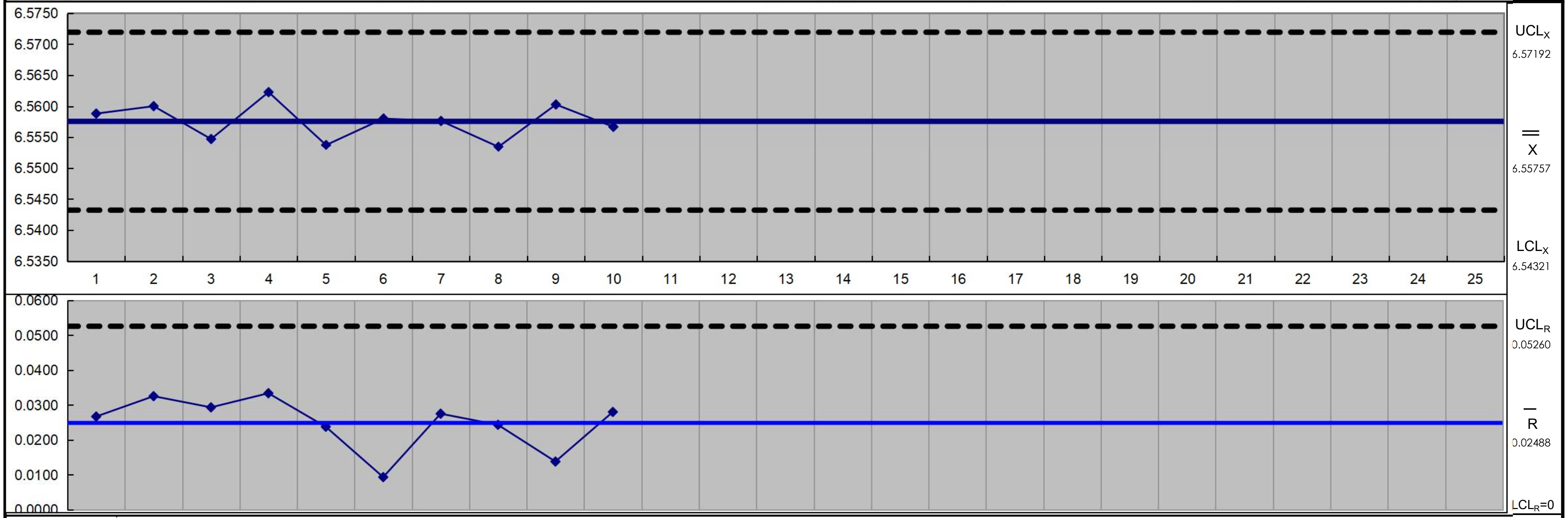


chart analysis	a)Note：	R chart and X chart should be analysed seperately, R chart first，then X chart.	b) out of limit：	point out of upper or lower limit, include point on the limit line.
	c)Chain：	1> 7 continuous points in one side of average line.	d) clane：	7 continuous points with trend, go up or go down.
	e)Close：	1> about 2/3 points in 1/3 limits lines area. 2> about 2/3 points close to average line.		

unnormal point analyse and improvement method

CHING MING METAL & TECHNOLOGY CO.,LTD

NO: 20250721001

DATE：25-7-21

spec	6.6	
upper tol.	0.2	6.800
lower tol.	-0.1	6.500

CUSTOMER: UAES
Part Name: Throttle Housing
Part NO:F01R 20Y 574

Instrument: C
Inspector:Liu yajuan
Drawing item:267-3#

CP= 5.685
Ca= -0.681
Validation:

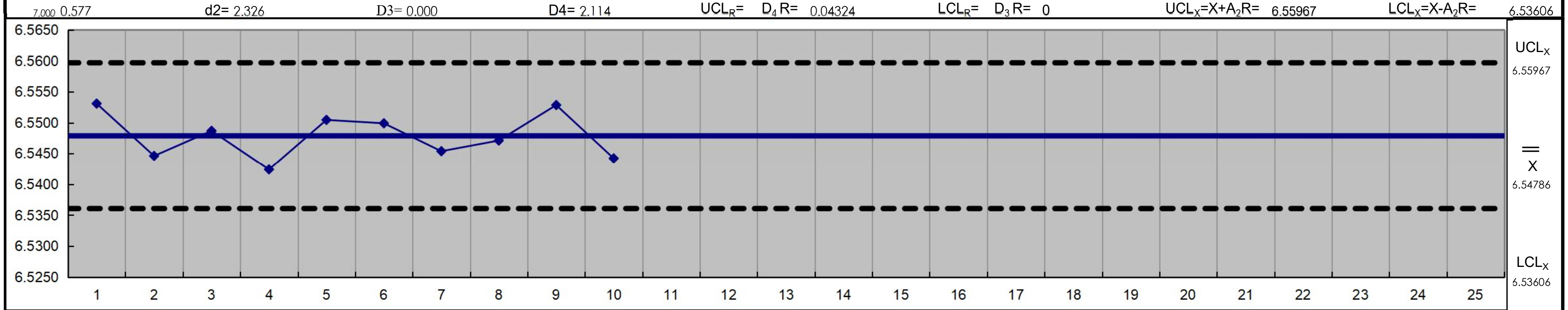
CPK= 1.814
PP= 5.497
Approval：

PpK= 1.754

Tmax-Tmin	0.0205	Min	6.53	average	6.55
		Max	6.56	σR	0.01
				σS	0.01

team	No. 1	No. 2	No. 3	No. 4	No. 5	No. 6	No. 7	No. 8	No. 9	No. 10	No. 11	No. 12	No. 13	No. 14	No. 15	No. 16	No. 17	No. 18	No. 19	No. 20	No. 21	No. 22	No. 23	No. 24	No. 25	average
sample 1	6.54	6.55	6.56	6.54	6.56	6.56	6.55	6.55	6.56	6.53																6.55
sample 2	6.56	6.55	6.56	6.56	6.55	6.54	6.55	6.54	6.54	6.53																6.55
sample 3	6.55	6.55	6.54	6.54	6.55	6.56	6.55	6.55	6.56	6.56																6.55
sample 4	6.56	6.54	6.54	6.54	6.53	6.55	6.54	6.55	6.56	6.54																6.55
sample 5	6.56	6.53	6.55	6.54	6.56	6.54	6.54	6.54	6.54	6.56																6.55
sample 6	6.553	6.545	6.549	6.542	6.550	6.550	6.545	6.547	6.553	6.544																6.548
sample 7	0.017	0.018	0.023	0.026	0.028	0.018	0.015	0.011	0.021	0.028																0.020

7.000 0.577　d2= 2.326　D3= 0.000　D4= 2.114　$UCL_R = D_4\bar{R} =$ 0.04324　$LCL_R = D_3\bar{R} =$ 0　$UCL_X = \bar{\bar{X}} + A_2\bar{R} =$ 6.55967　$LCL_X = \bar{\bar{X}} - A_2\bar{R} =$ 6.53606



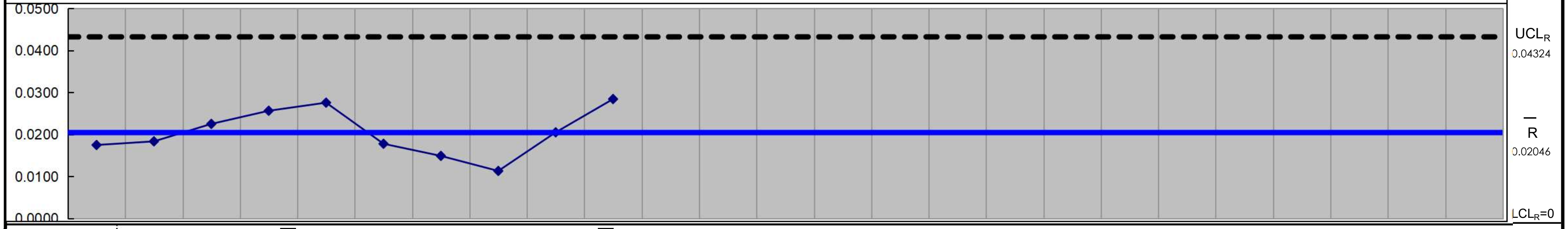


chart analysis	a)Note：	R chart and X chart should be analysed seperately, R chart first，then X chart.	b) out of limit：	point out of upper or lower limit, include point on the limit line.
	c)Chain：	1> 7 continuous points in one side of average line.	d) clane：	7 continuous points with trend, go up or go down.
	e)Close：	1> about 2/3 points in 1/3 limits lines area. 2> about 2/3 points close to average line.		

unnormal point analyse and improvement method

JM/QR-321/REV:0/Sep:01.2005

精明（西安）金属科技有限公司

CHING MING METAL & TECHNOLOGY CO.,LTD

NO: 20250721001

DATE：25-7-21

spec	6.6	
upper tol.	0.2	6.800
lower tol.	-0.1	6.500

CUSTOMER: UAES
Part Name: Throttle Housing
Part NO:F01R 20Y 574

Instrument: C
Inspector:Liu yajuan
Drawing item:267-4#

CP= 4.972
Ca= -0.135
Validation:

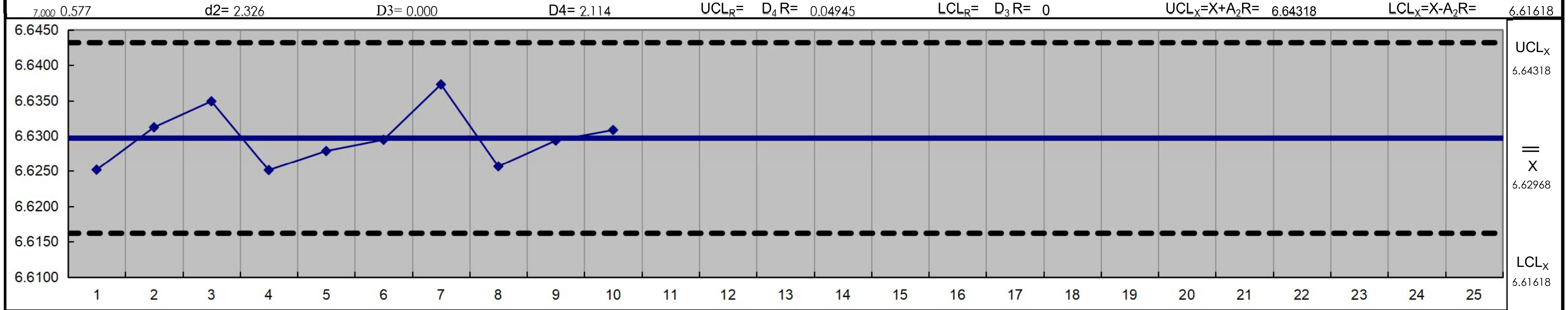
CPK= 4.299
PP= 5.260
Approval：

PpK= 4.547

				σ_S	0.01
		Max	6.65	σ_R	0.01
Tmax-Tmin	0.0234	Min	6.61	average	6.63

team	No. 1	No. 2	No. 3	No. 4	No. 5	No. 6	No. 7	No. 8	No. 9	No. 10	No. 11	No. 12	No. 13	No. 14	No. 15	No. 16	No. 17	No. 18	No. 19	No. 20	No. 21	No. 22	No. 23	No. 24	No. 25	average
sample 1	6.61	6.62	6.64	6.63	6.61	6.61	6.64	6.63	6.62	6.64																6.63
sample 2	6.62	6.64	6.62	6.62	6.63	6.65	6.62	6.62	6.65	6.64																6.63
sample 3	6.64	6.62	6.64	6.63	6.62	6.63	6.65	6.63	6.63	6.61																6.63
sample 4	6.63	6.63	6.63	6.62	6.63	6.63	6.64	6.63	6.63	6.64																6.63
sample 5	6.63	6.64	6.64	6.63	6.64	6.63	6.64	6.62	6.61	6.63																6.63
sample 6	6.625	6.631	6.635	6.625	6.628	6.629	6.637	6.626	6.629	6.631																6.630
sample 7	0.021	0.013	0.025	0.009	0.028	0.036	0.026	0.017	0.033	0.025																0.023

7.000 0.577 d2= 2.326 D3= 0.000 D4= 2.114 $UCL_R = D_4\bar{R} =$ 0.04945 $LCL_R = D_3\bar{R} =$ 0 $UCL_X = \bar{\bar{X}} + A_2\bar{R} =$ 6.64318 $LCL_X = \bar{\bar{X}} - A_2\bar{R} =$ 6.61618



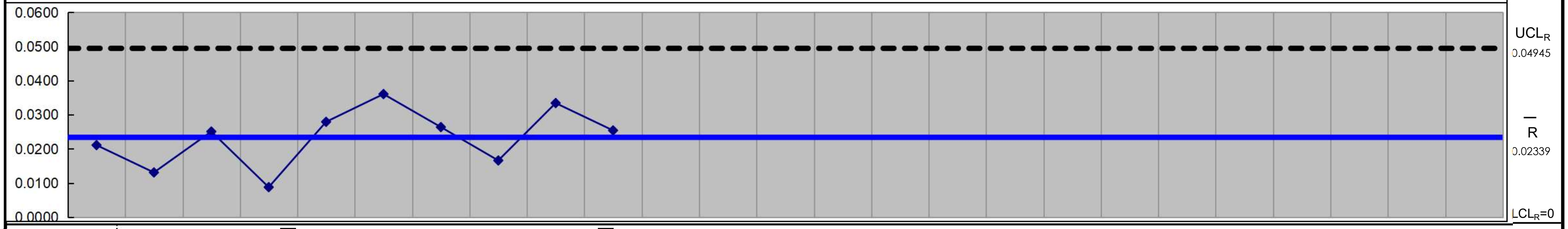


chart analysis	a)Note：	R chart and X chart should be analysed seperately, R chart first，then X chart.
	b) out of limit：	point out of upper or lower limit, include point on the limit line.
	c)Chain：	1> 7 continuous points in one side of average line.
	d) clane：	7 continuous points with trend, go up or go down.
	e)Close：	1> about 2/3 points in 1/3 limits lines area. 2> about 2/3 points close to average line.
unnormal point analyse and improvement method		

CHING MING METAL & TECHNOLOGY CO.,LTD

NO: 20250721001

DATE：25-7-21

spec	6.6	
upper tol.	0.2	6.800
lower tol.	-0.1	6.500

CUSTOMER: UAES
Part Name: Throttle Housing
Part NO:F01R 20Y 574

Instrument: C
Inspector:Liu yajuan
Drawing item:267-5#

CP= 3.734
Ca= -0.255
Validation:

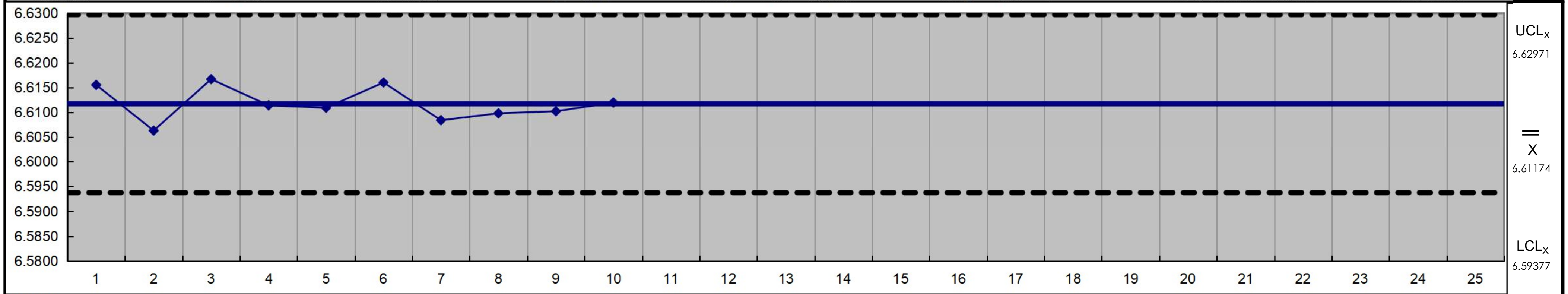
CPK= 2.782
PP= 3.986
Approval：

PpK= 2.969

		Max	6.63	σ_S	0.01
				σ_R	0.01
Tmax-Tmin	0.0311	Min	6.59	average	6.61

team	No. 1	No. 2	No. 3	No. 4	No. 5	No. 6	No. 7	No. 8	No. 9	No. 10	No. 11	No. 12	No. 13	No. 14	No. 15	No. 16	No. 17	No. 18	No. 19	No. 20	No. 21	No. 22	No. 23	No. 24	No. 25	average
sample 1	6.61	6.62	6.63	6.61	6.63	6.61	6.63	6.59	6.59	6.59																6.61
sample 2	6.63	6.63	6.61	6.62	6.59	6.63	6.60	6.61	6.61	6.62																6.61
sample 3	6.62	6.59	6.61	6.62	6.61	6.62	6.62	6.63	6.60	6.60																6.61
sample 4	6.60	6.59	6.60	6.61	6.61	6.61	6.59	6.60	6.63	6.63																6.61
sample 5	6.62	6.60	6.63	6.60	6.62	6.61	6.60	6.63	6.62	6.61																6.61
sample 6	6.616	6.606	6.617	6.611	6.611	6.616	6.608	6.610	6.610	6.612																6.612
sample 7	0.024	0.033	0.026	0.026	0.033	0.022	0.036	0.036	0.037	0.038																0.031

7.000 0.577 d2= 2.326 D3= 0.000 D4= 2.114 UCL_R= $D_4\bar{R}$= 0.06584 LCL_R= $D_3\bar{R}$= 0 $UCL_X=\bar{\bar{X}}+A_2\bar{R}$= 6.62971 $LCL_X=\bar{\bar{X}}-A_2\bar{R}$= 6.59377



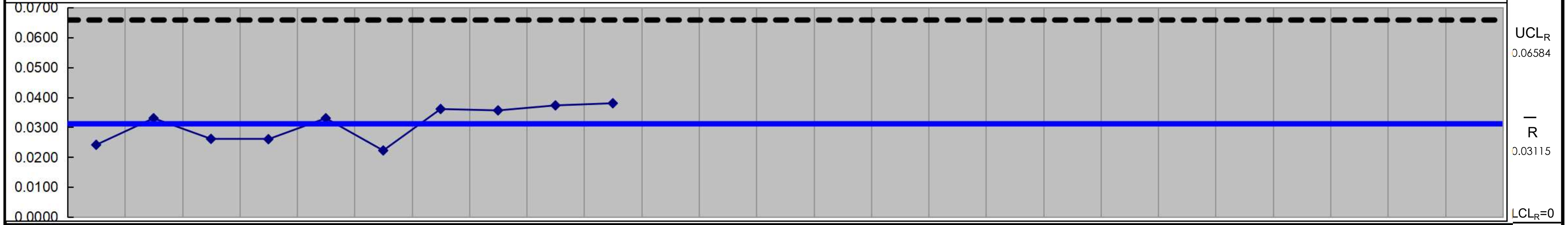


chart analysis	a)Note： R chart and X chart should be analysed seperately, R chart first，then X chart. b) out of limit： point out of upper or lower limit, include point on the limit line. c)Chain： 1> 7 continuous points in one side of average line. d) clane： 7 continuous points with trend, go up or go down. e)Close： 1> about 2/3 points in 1/3 limits lines area. 2> about 2/3 points close to average line.
unnormal point analyse and improvement method	

JM/QR-321/REV:0/Sep:01.2005

精明（西安）金属科技有限公司

CHING MING METAL & TECHNOLOGY CO.,LTD

NO: 20250721001 DATE：25-7-21

spec	6.6	
upper tol.	0.2	6.800
lower tol.	-0.1	6.500

CUSTOMER: UAES
Part Name: Throttle Housing
Part NO:F01R 20Y 574

Instrument: C
Inspector:Liu yajuan
Drawing item:267-6#

CP= 5.244
Ca= -0.408
Validation:

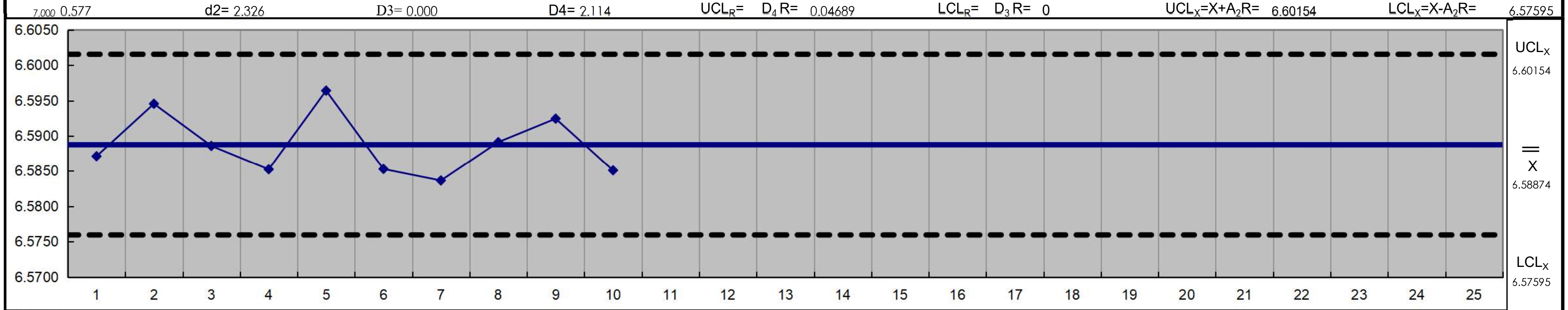
CPK= 3.102
PP= 4.978
Approval：

PpK= 2.945

				σ_S	0.01
		Max	6.61	σ_R	0.01
Tmax-Tmin	0.0222	Min	6.57	average	6.59

team	No. 1	No. 2	No. 3	No. 4	No. 5	No. 6	No. 7	No. 8	No. 9	No. 10	No. 11	No. 12	No. 13	No. 14	No. 15	No. 16	No. 17	No. 18	No. 19	No. 20	No. 21	No. 22	No. 23	No. 24	No. 25	average
sample 1	6.58	6.60	6.59	6.60	6.60	6.58	6.59	6.60	6.58	6.59																6.59
sample 2	6.58	6.60	6.59	6.60	6.58	6.59	6.57	6.59	6.60	6.59																6.59
sample 3	6.61	6.59	6.59	6.57	6.59	6.60	6.58	6.59	6.58	6.59																6.59
sample 4	6.60	6.59	6.59	6.58	6.60	6.58	6.59	6.59	6.61	6.57																6.59
sample 5	6.57	6.60	6.58	6.57	6.60	6.57	6.59	6.58	6.60	6.58																6.59
sample 6	6.587	6.595	6.589	6.585	6.596	6.585	6.584	6.589	6.592	6.585																6.589
sample 7	0.034	0.011	0.010	0.032	0.021	0.030	0.018	0.016	0.026	0.024																0.022

7.000 0.577 d2= 2.326 D3= 0.000 D4= 2.114 $UCL_R=$ $D_4\bar{R}=$ 0.04689 $LCL_R=$ $D_3\bar{R}=$ 0 $UCL_X=\bar{\bar{X}}+A_2\bar{R}=$ 6.60154 $LCL_X=\bar{\bar{X}}-A_2\bar{R}=$ 6.57595



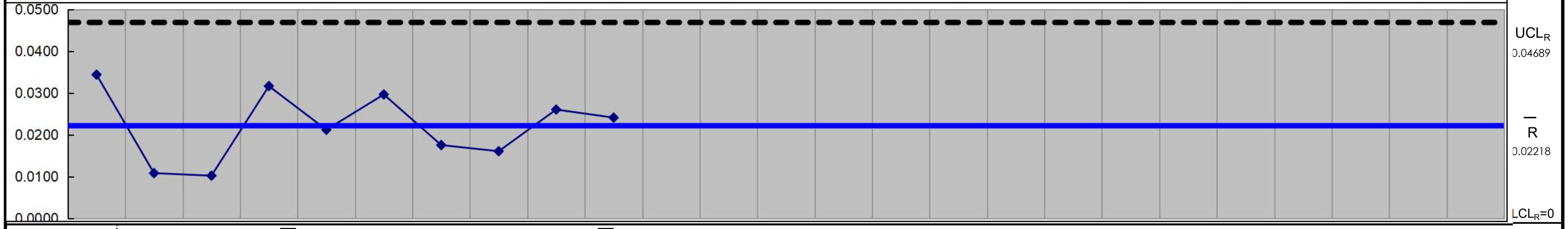


chart analysis		
	a)Note：	R chart and X chart should be analysed seperately, R chart first，then X chart.
	b) out of limit：	point out of upper or lower limit, include point on the limit line.
	c)Chain：	1> 7 continuous points in one side of average line.
	d) clane：	7 continuous points with trend, go up or go down.
	e)Close：	1> about 2/3 points in 1/3 limits lines area. 2> about 2/3 points close to average line.

unnormal point analyse and improvement method

JM/QR-321/REV:0/Sep:01.2005

精明（西安）金属科技有限公司